

Switchdev BoF

netdev 1.2

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Agenda

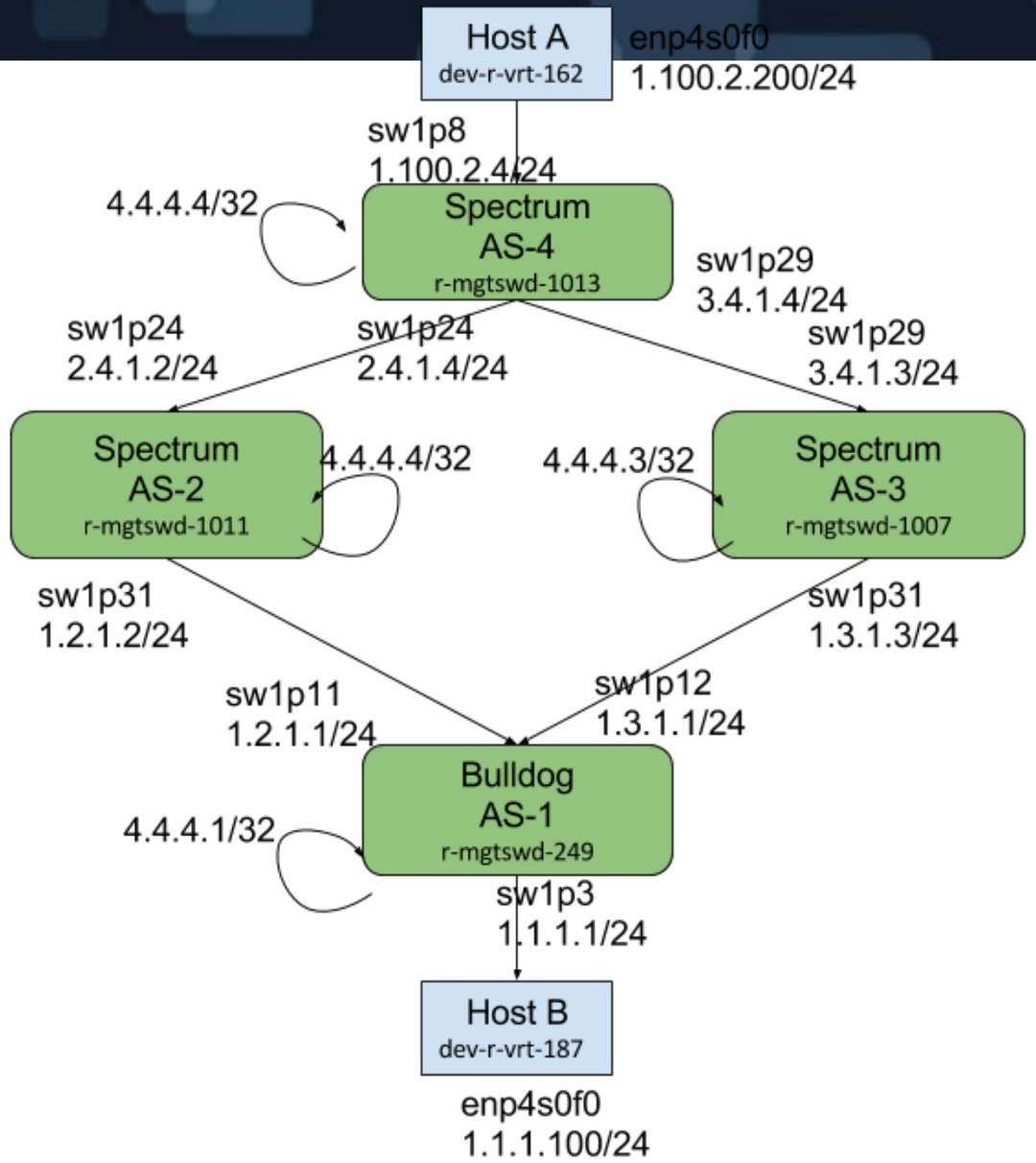
- Status update
- Routing offload abort, how to fix it
- Switchdev visibility & debugability
- Ordering Problems in Hardware Offloading
- CPU port Hardware Offloading

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OSPF

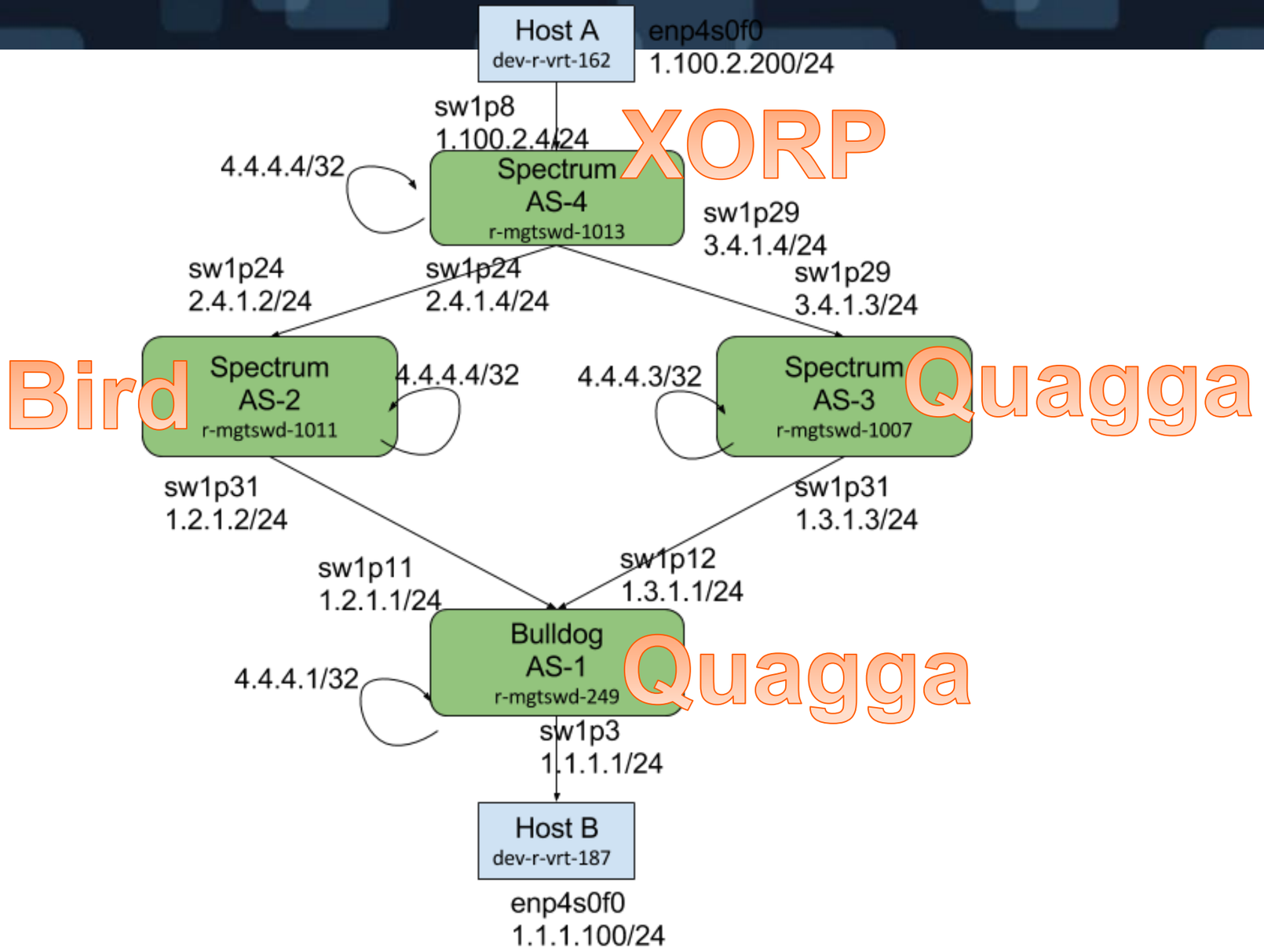
Quagga, XORP, Bird

Demo



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Demo



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Configuration OSPFv2

249 quagga.conf

```
router ospf
  network 1.1.1.0/24 area 0
  network 1.2.1.0/24 area 0
  network 1.3.1.0/24 area 0
  network 4.4.4.1/32 area 0
```

1011 bird.conf

```
protocol ospf mlxsw_ospf {
  tick 2;
  rfc1583compat yes;
  area 0 {
    stub no;
    interface "sw1p31" {
      rx buffer large;
      poll 14;
      dead count 5;
      dead 40;
    };
    interface "sw1p24" {
      rx buffer large;
      poll 14;
      dead count 5;
      dead 40;
    };
    interface "lo" {
      rx buffer large;
      poll 14;
      dead count 5;
      dead 40;
    };
  };
}
```

1013 xorp.ospfv2.conf

```
protocols {
  ospf4 {
  router-id: 4.4.4.4
  area 0.0.0.0 {

  interface sw1p8 {
    vif sw1p8 {
      address 1.100.2.4 {
        router-dead-interval: 40
    }
    }
  }
  interface sw1p24 {
    vif sw1p24 {
      address 2.4.1.4 {
        router-dead-interval: 40
      }
    }
  }
  interface sw1p29 {
    vif sw1p29 {
      address 3.4.1.4 {
        router-dead-interval: 40
      }
      }
```

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Quagga

```
[root@r-mgtswd-249 ~]# ip route
default via 10.209.0.1 dev enp0s20f0
1.1.1.0/24 dev sw1p3  proto kernel  scope link  src 1.1.1.1 offload
1.2.1.0/24 dev sw1p11  proto kernel  scope link  src 1.2.1.1 offload
1.3.1.0/24 dev sw1p12  proto kernel  scope link  src 1.3.1.1 offload
1.100.2.0/24  proto zebra  metric 21 offload
        nexthop via 1.2.1.2  dev sw1p11 weight 1
        nexthop via 1.3.1.3  dev sw1p12 weight 1
2.4.1.0/24 via 1.2.1.2 dev sw1p11  proto zebra  metric 20 offload
3.4.1.0/24 via 1.3.1.3 dev sw1p12  proto zebra  metric 20 offload
4.4.4.2 via 1.2.1.2 dev sw1p11  proto zebra  metric 10 offload
4.4.4.3 via 1.3.1.3 dev sw1p12  proto zebra  metric 20 offload
10.209.0.0/23 dev enp0s20f0  proto kernel  scope link  src 10.209.0.231
169.254.0.0/16 dev enp0s20f0  scope link  metric 1004
[root@r-mgtswd-249 ~]# vtysh

Hello, this is Quagga (version 0.99.24.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

r-mgtswd-249.mtr.labs.mlnx# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, P - PIM, A - Babel,
       > - selected route, * - FIB route

K>* 0.0.0.0/0 via 10.209.0.1, enp0s20f0
O   1.1.1.0/24 [110/10] is directly connected, sw1p3, 1d05h20m
C>* 1.1.1.0/24 is directly connected, sw1p3
O   1.2.1.0/24 [110/10] is directly connected, sw1p11, 04:15:45
C>* 1.2.1.0/24 is directly connected, sw1p11
O   1.3.1.0/24 [110/10] is directly connected, sw1p12, 1d05h20m
C>* 1.3.1.0/24 is directly connected, sw1p12
O>* 1.100.2.0/24 [110/21] via 1.2.1.2, sw1p11, 00:00:36
  *                       via 1.3.1.3, sw1p12, 00:00:36
O>* 2.4.1.0/24 [110/20] via 1.2.1.2, sw1p11, 04:15:45
O>* 3.4.1.0/24 [110/20] via 1.3.1.3, sw1p12, 00:00:42
O   4.4.4.1/32 [110/10] is directly connected, lo, 1d05h20m
C>* 4.4.4.1/32 is directly connected, lo
O>* 4.4.4.2/32 [110/10] via 1.2.1.2, sw1p11, 04:15:45
O>* 4.4.4.3/32 [110/20] via 1.3.1.3, sw1p12, 00:00:42
C>* 10.209.0.0/23 is directly connected, enp0s20f0
C>* 127.0.0.0/8 is directly connected, lo
K>* 169.254.0.0/16 is directly connected, enp0s20f0
```

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Bird

```
[root@r-mgtswd-1011 eladr]# ip route
default via 10.209.0.1 dev eno1
1.1.1.0/24 via 1.2.1.1 dev sw1p31  proto bird offload
1.2.1.0/24 dev sw1p31  proto kernel  scope link  src 1.2.1.2 offload
1.3.1.0/24 via 1.2.1.1 dev sw1p31  proto bird offload
1.100.2.0/24 via 2.4.1.4 dev sw1p24  proto bird offload
2.4.1.0/24 dev sw1p24  proto kernel  scope link  src 2.4.1.2 offload
3.4.1.0/24 via 2.4.1.4 dev sw1p24  proto bird offload
4.4.4.1 via 1.2.1.1 dev sw1p31  proto bird offload
4.4.4.3 via 2.4.1.4 dev sw1p24  proto bird offload
10.209.0.0/23 dev eno1  proto kernel  scope link  src 10.209.0.60
169.254.0.0/16 dev eno1  scope link  metric 1004
[root@r-mgtswd-1011 eladr]# birdcl
BIRD 1.6.0 ready.
bird> show route
1.1.1.0/24         via 1.2.1.1 on sw1p31 [mlxsw_ospf 11:55:29] * I (150/20) [4.4.4.1]
1.2.1.0/24         dev sw1p31 [mlxsw_ospf 11:45:10] * I (150/10) [10.209.0.60]
1.3.1.0/24         via 1.2.1.1 on sw1p31 [mlxsw_ospf 11:55:29] * I (150/20) [4.4.4.1]
3.4.1.0/24         via 2.4.1.4 on sw1p24 [mlxsw_ospf 16:09:49] * I (150/11) [4.4.4.4]
2.4.1.0/24         dev sw1p24 [mlxsw_ospf 11:45:10] * I (150/10) [10.209.0.60]
4.4.4.1/32         via 1.2.1.1 on sw1p31 [mlxsw_ospf 11:55:29] * I (150/20) [4.4.4.1]
4.4.4.2/32         dev lo [mlxsw_ospf 11:45:10] * I (150/0) [10.209.0.60]
4.4.4.3/32         via 2.4.1.4 on sw1p24 [mlxsw_ospf 16:10:29] * I (150/21) [4.4.4.3]
1.100.2.0/24       via 2.4.1.4 on sw1p24 [mlxsw_ospf 15:56:29] * I (150/11) [4.4.4.4]
bird>
```

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XORP

```
[root@r-mgtswd-1013 xorp]# ip route
default via 10.209.0.1 dev eno1
1.1.1.0/24 via 2.4.1.2 dev sw1p24  proto xorp  metric 21 offload notify
1.2.1.0/24 via 2.4.1.2 dev sw1p24  proto xorp  metric 11 offload notify
1.3.1.0/24 via 3.4.1.3 dev sw1p29  proto xorp  metric 11 offload notify
1.100.2.0/24 dev sw1p8  proto kernel  scope link  src 1.100.2.4 offload
2.4.1.0/24 dev sw1p24  proto kernel  scope link  src 2.4.1.4 offload
3.4.1.0/24 dev sw1p29  proto kernel  scope link  src 3.4.1.4 offload
4.4.4.1 via 2.4.1.2 dev sw1p24  proto xorp  metric 21 offload notify
4.4.4.2 via 2.4.1.2 dev sw1p24  proto xorp  metric 1 offload notify
4.4.4.3 via 3.4.1.3 dev sw1p29  proto xorp  metric 11 offload notify
10.209.0.0/23 dev eno1  proto kernel  scope link  src 10.209.0.62
169.254.0.0/16 dev eno1  scope link  metric 1004
[root@r-mgtswd-1013 xorp]# xorpsh
Welcome to XORP v1.8.6-WIP on r-mgtswd-1013.mtr.labs.mlnx
Version tag: 00000000  Build Date: 2016-07-28 11:52 64-bit
root@r-mgtswd-1013.mtr.labs.mlnx> show route table ipv4 unicast ospf
1.1.1.0/24      [ospf(110)/21]
                > to 2.4.1.2 via sw1p24/sw1p24
1.2.1.0/24      [ospf(110)/11]
                > to 2.4.1.2 via sw1p24/sw1p24
1.3.1.0/24      [ospf(110)/11]
                > to 3.4.1.3 via sw1p29/sw1p29
4.4.4.1/32      [ospf(110)/21]
                > to 2.4.1.2 via sw1p24/sw1p24
4.4.4.2/32      [ospf(110)/1]
                > to 2.4.1.2 via sw1p24/sw1p24
4.4.4.3/32      [ospf(110)/11]
                > to 3.4.1.3 via sw1p29/sw1p29
root@r-mgtswd-1013.mtr.labs.mlnx>
```

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Traffic

```
C>* 4.4.4.1/32 is directly connected, lo
O>* 4.4.4.2/32 [110/10] via 1.2.1.2, sw1p11, 04:15:45
O>* 4.4.4.3/32 [110/20] via 1.3.1.3, sw1p12, 00:00:42
C>* 10.209.0.0/23 is directly connected, enp0s20f0
C>* 127.0.0.0/8 is directly connected, lo
K>* 169.254.0.0/16 is directly connected, enp0s20f0
r-mgtswd-249.mtr.labs.mlnx#
r-mgtswd-249.mtr.labs.mlnx# exit
[root@r-mgtswd-249 ~]# ip route
default via 10.209.0.1 dev enp0s20f0
1.1.1.0/24 dev sw1p3  proto kernel  scope link  src 1.1.1.1
 offload
1.2.1.0/24 dev sw1p11  proto kernel  scope link  src 1.2.1.
1 offload
1.3.1.0/24 dev sw1p12  proto kernel  scope link  src 1.3.1.
1 offload
1.100.2.0/24  proto zebra  metric 21 offload
        nexthop via 1.2.1.2  dev sw1p11 weight 1
        nexthop via 1.3.1.3  dev sw1p12 weight 1
2.4.1.0/24 via 1.2.1.2 dev sw1p11  proto zebra  metric 20 o
ffload
3.4.1.0/24 via 1.3.1.3 dev sw1p12  proto zebra  metric 20 o
ffload
4.4.4.2 via 1.2.1.2 dev sw1p11  proto zebra  metric 10 offl
oad
4.4.4.3 via 1.3.1.3 dev sw1p12  proto zebra  metric 20 offl
oad
10.209.0.0/23 dev enp0s20f0  proto kernel  scope link  src
10.209.0.231
169.254.0.0/16 dev enp0s20f0  scope link  metric 1004
[root@r-mgtswd-249 ~]#
```

```
2.4.1.0/24 dev sw1p24  proto kernel  scope link  src 2.4.1
.2 offload
3.4.1.0/24 via 2.4.1.4 dev sw1p24  proto bird offload
4.4.4.1 via 1.2.1.1 dev sw1p31  proto bird offload
4.4.4.3 via 2.4.1.4 dev sw1p24  proto bird offload
10.209.0.0/23 dev eno1  proto kernel  scope link  src 10.2
09.0.60
169.254.0.0/16 dev eno1  scope link  metric 1004
[root@r-mgtswd-1011 eladr]# birdcl
BIRD 1.6.0 ready.
bird> show route
1.1.1.0/24         via 1.2.1.1 on sw1p31 [mlxsw_ospf 11:55
:29] * I (150/20) [4.4.4.1]
1.2.1.0/24         dev sw1p31 [mlxsw_ospf 11:45:10] * I (1
50/10) [10.209.0.60]
1.3.1.0/24         via 1.2.1.1 on sw1p31 [mlxsw_ospf 11:55
:29] * I (150/20) [4.4.4.1]
3.4.1.0/24         via 2.4.1.4 on sw1p24 [mlxsw_ospf 16:09
:49] * I (150/11) [4.4.4.4]
2.4.1.0/24         dev sw1p24 [mlxsw_ospf 11:45:10] * I (1
50/10) [10.209.0.60]
4.4.4.1/32         via 1.2.1.1 on sw1p31 [mlxsw_ospf 11:55
:29] * I (150/20) [4.4.4.1]
4.4.4.2/32         dev lo [mlxsw_ospf 11:45:10] * I (150/0
) [10.209.0.60]
4.4.4.3/32         via 2.4.1.4 on sw1p24 [mlxsw_ospf 16:10
:29] * I (150/21) [4.4.4.3]
1.100.2.0/24       via 2.4.1.4 on sw1p24 [mlxsw_ospf 15:56
:29] * I (150/11) [4.4.4.4]
bird>
bird>
```

```
[root@r-mgtswd-1007 quagga]# packet_write_wait: Connection
to 10.209.1.6: Broken pipe
[eladr@dev-r-vrt-157-005 ~]$ ssh r-mgtswd-1007
Last login: Wed Jul 27 09:40:11 2016 from 10.212.157.5
[eladr@r-mgtswd-1007 ~]$ sudo su
[root@r-mgtswd-1007 eladr]# ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state
 UNKNOWN mode DEFAULT group default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: sit0@NONE: <NOARP> mtu 1480 qdisc noop state DOWN mode D
EFAULT group default qlen 1
    link/sit 0.0.0.0 brd 0.0.0.0
3: ip6tnl0@NONE: <NOARP> mtu 1452 qdisc noop state DOWN mod
e DEFAULT group default qlen 1
    link/tunnel6 :: brd ::
4: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc f
q_codel state UP mode DEFAULT group default qlen 1000
    link/ether e4:1d:2d:f7:82:d6 brd ff:ff:ff:ff:ff:ff
5: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DO
WN mode DEFAULT group default qlen 1000
    link/ether e4:1d:2d:f7:82:d7 brd ff:ff:ff:ff:ff:ff
[root@r-mgtswd-1007 eladr]# modprobe mlxsw_pci
[root@r-mgtswd-1007 eladr]# systemctl start zebra
[root@r-mgtswd-1007 eladr]# systemctl start ospfd
[root@r-mgtswd-1007 eladr]#
```

```
.4 offload
3.4.1.0/24 dev sw1p29  proto kernel  scope link  src 3.4.1
.4 offload
4.4.4.1 via 2.4.1.2 dev sw1p24  proto xorp  metric 21 offl
oad notify
4.4.4.2 via 2.4.1.2 dev sw1p24  proto xorp  metric 1 offlo
ad notify
4.4.4.3 via 3.4.1.3 dev sw1p29  proto xorp  metric 11 offl
oad notify
10.209.0.0/23 dev eno1  proto kernel  scope link  src 10.2
09.0.62
169.254.0.0/16 dev eno1  scope link  metric 1004
[root@r-mgtswd-1013 xorp]# xorpsh
Welcome to XORP v1.8.6-WIP on r-mgtswd-1013.mtr.labs.mlnx
Version tag: 00000000  Build Date: 2016-07-28 11:52 64-bit
root@r-mgtswd-1013.mtr.labs.mlnx> show route table ipv4 un
icast ospf
1.1.1.0/24      [ospf(110)/21]
                > to 2.4.1.2 via sw1p24/sw1p24
1.2.1.0/24      [ospf(110)/11]
                > to 2.4.1.2 via sw1p24/sw1p24
1.3.1.0/24      [ospf(110)/11]
                > to 3.4.1.3 via sw1p29/sw1p29
4.4.4.1/32      [ospf(110)/21]
                > to 2.4.1.2 via sw1p24/sw1p24
4.4.4.2/32      [ospf(110)/1]
                > to 2.4.1.2 via sw1p24/sw1p24
4.4.4.3/32      [ospf(110)/11]
                > to 3.4.1.3 via sw1p29/sw1p29
root@r-mgtswd-1013.mtr.labs.mlnx>
root@r-mgtswd-1013.mtr.labs.mlnx>
```

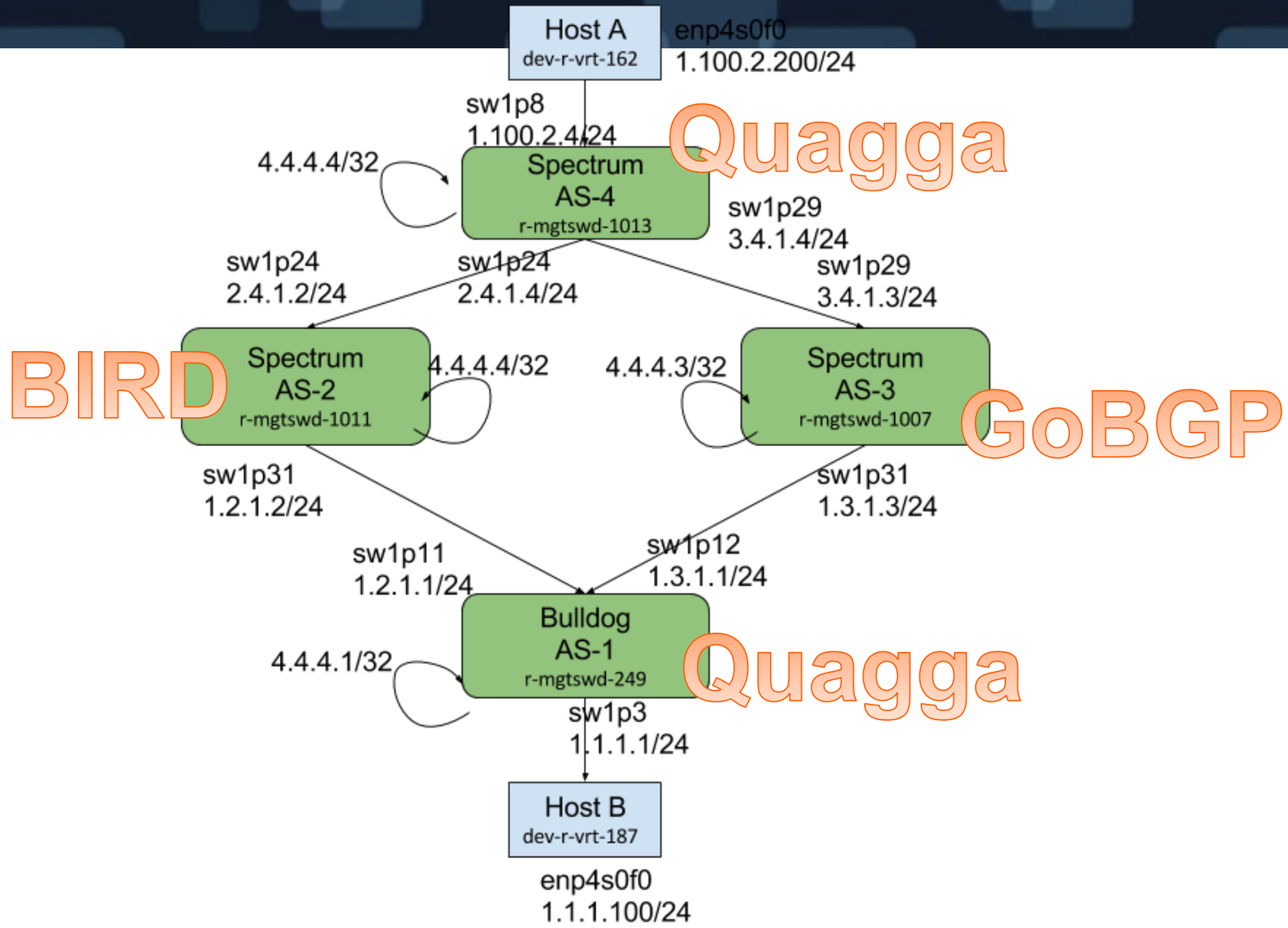
```
[  4]  7.0- 8.0 sec   384 MBytes  3.22 Gbits/sec
[SUM]  7.0- 8.0 sec  3.08 GBytes  26.4 Gbits/sec
[  9]  8.0- 9.0 sec   440 MBytes  3.69 Gbits/sec
[ 10]  8.0- 9.0 sec   470 MBytes  3.94 Gbits/sec
[  6]  8.0- 9.0 sec   437 MBytes  3.67 Gbits/sec
[  4]  8.0- 9.0 sec   436 MBytes  3.66 Gbits/sec
[  3]  8.0- 9.0 sec   480 MBytes  4.02 Gbits/sec
[  5]  8.0- 9.0 sec   455 MBytes  3.82 Gbits/sec
[  7]  8.0- 9.0 sec   260 MBytes  2.18 Gbits/sec
[  8]  8.0- 9.0 sec   231 MBytes  1.93 Gbits/sec
[SUM]  8.0- 9.0 sec  3.13 GBytes  26.9 Gbits/sec
[  9]  9.0-10.0 sec   433 MBytes  3.63 Gbits/sec
[  9]  0.0-10.0 sec  4.10 GBytes  3.52 Gbits/sec
[ 10]  9.0-10.0 sec   625 MBytes  5.25 Gbits/sec
[ 10]  0.0-10.0 sec  4.59 GBytes  3.95 Gbits/sec
[  5]  9.0-10.0 sec   344 MBytes  2.89 Gbits/sec
[  5]  0.0-10.0 sec  4.83 GBytes  4.15 Gbits/sec
[  4]  9.0-10.0 sec   455 MBytes  3.82 Gbits/sec
[  4]  0.0-10.0 sec  4.06 GBytes  3.49 Gbits/sec
[  7]  9.0-10.0 sec   401 MBytes  3.37 Gbits/sec
[  7]  0.0-10.0 sec  2.43 GBytes  2.09 Gbits/sec
[  8]  9.0-10.0 sec   402 MBytes  3.37 Gbits/sec
[  8]  0.0-10.0 sec  3.58 GBytes  3.07 Gbits/sec
[  6]  9.0-10.0 sec   333 MBytes  2.79 Gbits/sec
[  6]  0.0-10.1 sec  3.37 GBytes  2.85 Gbits/sec
[  3]  9.0-10.0 sec   206 MBytes  1.73 Gbits/sec
[SUM]  9.0-10.0 sec  3.13 GBytes  26.9 Gbits/sec
[  3]  0.0-10.2 sec  4.20 GBytes  3.54 Gbits/sec
[SUM]  0.0-10.2 sec  31.2 GBytes  26.3 Gbits/sec
[root@dev-r-vrt-187 ~]#
```

```
[  6]  7.0- 8.0 sec   384 MBytes  3.22 Gbits/sec
[SUM]  7.0- 8.0 sec  3.08 GBytes  26.4 Gbits/sec
[  4]  8.0- 9.0 sec   479 MBytes  4.02 Gbits/sec
[  5]  8.0- 9.0 sec   470 MBytes  3.94 Gbits/sec
[  7]  8.0- 9.0 sec   454 MBytes  3.81 Gbits/sec
[  9]  8.0- 9.0 sec   439 MBytes  3.68 Gbits/sec
[  6]  8.0- 9.0 sec   439 MBytes  3.69 Gbits/sec
[ 11]  8.0- 9.0 sec   438 MBytes  3.67 Gbits/sec
[ 10]  8.0- 9.0 sec   260 MBytes  2.18 Gbits/sec
[  8]  8.0- 9.0 sec   230 MBytes  1.93 Gbits/sec
[SUM]  8.0- 9.0 sec  3.13 GBytes  26.9 Gbits/sec
[  5]  9.0-10.0 sec   627 MBytes  5.26 Gbits/sec
[  7]  9.0-10.0 sec   346 MBytes  2.90 Gbits/sec
[  8]  9.0-10.0 sec   402 MBytes  3.37 Gbits/sec
[  6]  9.0-10.0 sec   454 MBytes  3.81 Gbits/sec
[ 11]  9.0-10.0 sec   434 MBytes  3.64 Gbits/sec
[ 10]  9.0-10.0 sec   401 MBytes  3.37 Gbits/sec
[  5]  0.0-10.0 sec  4.59 GBytes  3.95 Gbits/sec
[  7]  0.0-10.0 sec  4.83 GBytes  4.15 Gbits/sec
[  8]  0.0-10.0 sec  3.58 GBytes  3.07 Gbits/sec
[  6]  0.0-10.0 sec  4.06 GBytes  3.49 Gbits/sec
[ 11]  0.0-10.0 sec  4.10 GBytes  3.52 Gbits/sec
[ 10]  0.0-10.0 sec  2.43 GBytes  2.09 Gbits/sec
[  4]  9.0-10.0 sec   206 MBytes  1.73 Gbits/sec
[  9]  9.0-10.0 sec   331 MBytes  2.77 Gbits/sec
[SUM]  9.0-10.0 sec  3.13 GBytes  26.8 Gbits/sec
[  9]  0.0-10.2 sec  3.37 GBytes  2.85 Gbits/sec
[  4]  0.0-10.2 sec  4.20 GBytes  3.54 Gbits/sec
[SUM]  0.0-10.2 sec  31.2 GBytes  26.3 Gbits/sec
[root@dev-r-vrt-162-006 ~]#
```

BGP over OSPF loopbacks

Quagga, XORP, Bird, GoBGP

Demo



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Configuration BGP

249 quagga.conf

```
router bgp 1
          bgp router-id 10.0.0.1
          bgp bestpath as-path
multipath-relax
          bgp fast-external-failover
          maximum-paths 32
          bgp scan-time 5
          network 1.1.1.0/24
          neighbor 1.3.1.3 remote-as 3
          neighbor 1.2.1.2 remote-as 2
```

1011 bird.conf

```
protocol ospf mlxsw_ospf {
  tick 2;
  rfc1583compat yes;
  area 0 {
    stub no;
    interface "sw1p31" {
      rx buffer large;
      poll 14;
      dead count 5;
      dead 40;
    };
    interface "sw1p24" {
      rx buffer large;
      poll 14;
      dead count 5;
      dead 40;
    };
    interface "lo" {
      rx buffer large;
      poll 14;
      dead count 5;
      dead 40;
    };
  };
}
```

1007 GoBGP

```
[global.config]
  as = 3
  router-id = "10.0.0.3"

[[neighbors]]
  [neighbors.config]
    neighbor-address = "1.3.1.1"
    peer-as = 1

[[neighbors]]
  [neighbors.config]
    neighbor-address = "3.4.1.4"
    peer-as = 4

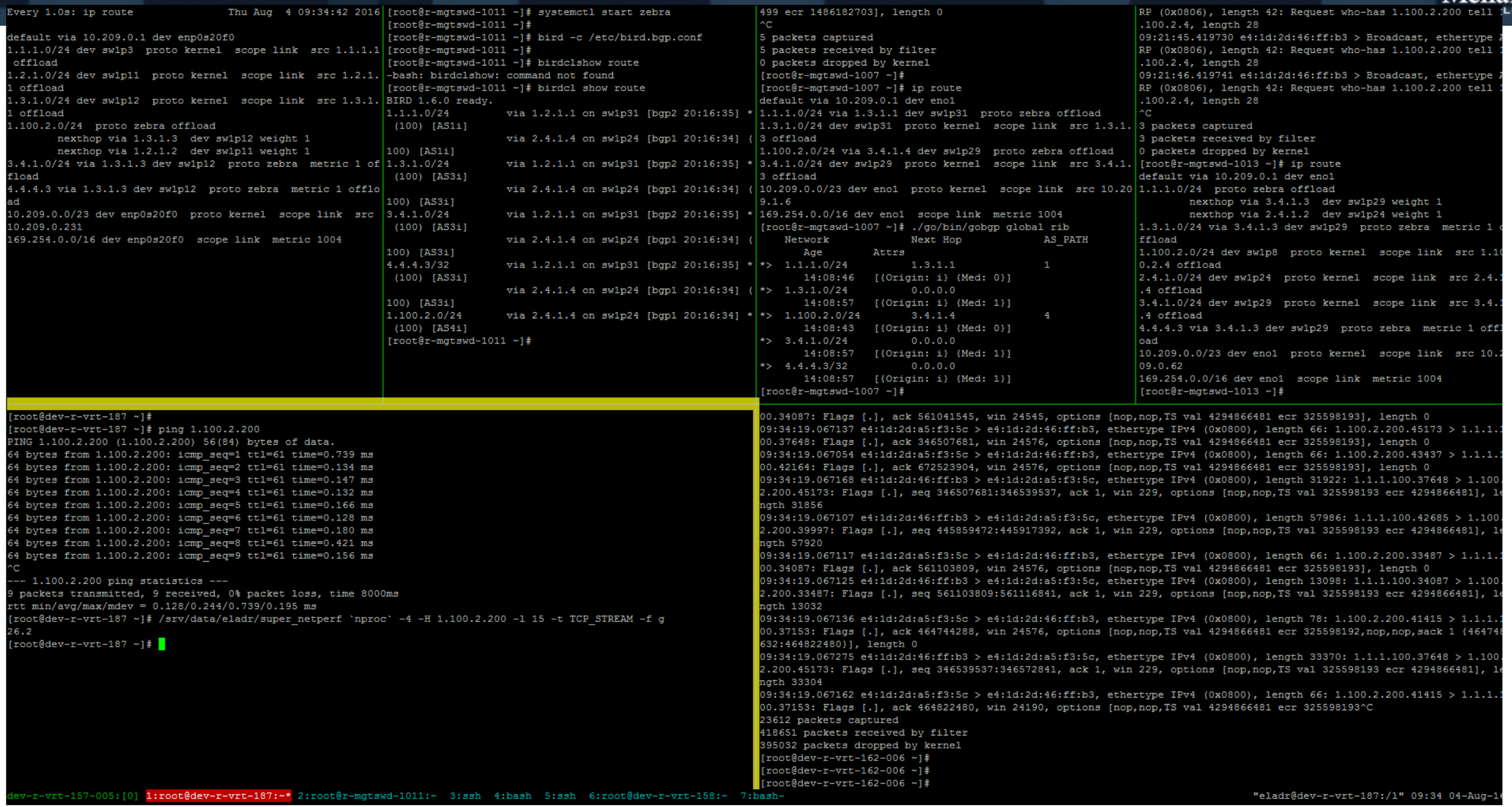
[zebra]
    [zebra.config]
        enabled = true
        url = "unix:/var/run/quagga/zserv.api"
        redistribute-route-type-list = ["connect"]
```

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Video

Video

Traffic with BGP





Routing offload abort, how to fix it

Switchdev BOF, Netdev 1.2,

Jiri Pirko j, matty Kadosh

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Initial FIB4 offload implementation

- switchdev add/del functions are called from FIB code for every added/deleted FIB entry
  - A lookup is made in upper-lower tree for netdevice which implements switchdev callbacks
  - netdevs to call FIB add/dev switchdev op are selected according to FIB entry nexthop netdev
    - Only routes related to the switch port netdevices get offloaded. The rest of the routes hardware has no clue about which leads to broken routing!! (issue #1)
      - `ip addr add 192.168.1.1/24 dev dummy0`
- If offload fails on any netdevice, abort mechanism is triggered
  - May happen in multiple cases:
    - FIB entry failed to be offloaded in specific driver, for example for capacity reasons
    - User executes “ip rule” to add a custom rule
  - FIB code calls switchdev abort function, then external flush FIB function is called
    - From that point, all traffic flows through kernel!! (issue #2)
      - In case of Spectrum ASIC, instead of 6.4Tb/s and 4.7B packets per second, the system delivers something around 4Gb/s (~1500x less, for big packets)
  - All offloaded entries are removed from all devices (switchdev del function)
  - A global flag is set indicating not to offload any FIB entries, ever
    - User has to reboot the system in order to make offloading to work again!! (issue #3)
  - Abort is silent. User does not know that abort mechanism was triggered

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Introduction of FIB notifier

- Merged a week ago
- Instead of netdev-oriented switchdev op, push down FIB entries by a notifier
  - Replaces switchdev add/del function calls in FIB code
  - Everyone who is interested registers a handler to receive FIB entries add/delete event
- Allows driver to get information about all FIB entries added to kernel
  - Traps could be set in hardware to push matching packets to kernel for processing
  - Fixes issue #1
- Moves the abort duties down to drivers
  - Inability to offload a FIB entry by one driver, does not affect other drivers
  - No need to reboot system in order to get offloads working again, reinsertion of driver module is enough
  - Partially fixes issue #3
- Issue #2 still stands
  - after abort, all traffic still goes through kernel!

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How to fix the issue #2? - The passive way

- Leave it be as it is
- Pros:
  - User does and sees everything as he is used to
  - LPM gives consistent results in software and in hardware
- Cons:
  - Completely unusable in real life

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How to fix the issue #2? - Synced-fail way

- Have entries synced between kernel and hardware as well, only in case of route add fail in hardware, propagate the failure to user
- Pros:
  - User does not experience abort mechanism
  - LPM gives consistent results in software and in hardware
- Cons:
  - Route add fails even in case of non-offloaded system it would not
    - but user should be prepared for fail anyway (-ENOMEM for example)

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How to fix the issue #2? - Synced-fail-if-asked-to way

- Have entries synced between kernel and hardware as well, only in case of route add fail in hardware, propagate the failure to user - but only if user allows it by an “allow to fail” flag in route add mgs
- Pros:
  - User does not experience abort mechanism if he uses the new “allow to fail” flag
    - The default is still to abort
  - LPM gives consistent results in software and in hardware
- Cons:
  - Apps have to be teached to work with the new flag
- I think this is a good approach to start with now and see how that works

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How to fix the issue #2? - Unsynced, SKIP_HW/SKIP_SW

- Similar to TC, allow user to pass flags to skip insertion of a route in either software (kernel) or hardware
- Pros:
  - User does not experience abort mechanism if he does not want to
    - The default would still be to sync and abort if necessary
  - User can freely decide how to manage the FIB table within kernel and hardware
- Cons:
  - LPM gives inconsistent results in software and in hardware
    - But that applies to TC offload as well, and everyone is quite happy
  - Apps have to be teached to work with the new flags

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Switchdev visibility, debugability

netdev 1.2

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Problem Statement(issue #1)

Offload dependencies

- HW switch silicon differ in pipeline
- May have different offload dependencies -> entry can be offloaded only when all chain of dependencies met
- Example route offloading
- HW 1
  - Router entry dependency graph FIB ->ARP -> FBD
- HW 2
  - Router entry dependency graph FI ->ARP
- Switchdev users should be able to know when a forwarding entry was offloaded

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Problem Statement (issue#2)

Pipeline visibility , debugability

- HW switch silicon differ in pipeline
  - Offloading Linux pipeline to different HW silicones may end with different results
- In most cases there is more then one way for Offloading Linux pipeline into single switch silicon
  - Offload tune for performance vs scale vs resolution vs insertion rate …
- Switchdev users should be able to “understand” what they get
  - Scalability , performance , insertion rate …
- Switchdev users should be able to debug the network

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Problem Statement- ECMP example

- Linux view

  192.168.4.0/24

  nexthop via 192.168.175.151 dev port0 weight 1(ECMP_1)

  nexthop via 192.168.176.151 dev port1 weight 1(ECMP_2)

  nexthop via 192.168.177.151 dev port2 weight 1(ECMP_3)

- HW 1 - HW adjacency size (T2) must be ^2

  Table T1 (router)

  match : 192.168.4.0/24 action : go to table T2 index 1

  Table T2(adjacency)

  Match: index select {size 4 : ECMP_1*2,ECMP_2,ECMP_3} ->ECMP 1 get 50% for the traffic

- HW 2 -HW adjacency size (T2) must be ^2

  Table T1 (router)

  match : 192.168.4.0/24 action : go to table T2 index 1,size 16 -> ECMP size change may result in a massive route updates

  Table T2(adjacency)

  Match: index select {ECMP_1*6,ECMP_2*5,ECMP_3*5}

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Problem Statement- ECMP cont.

- HW 3

  match : 192.168.4.0/24 action : go to table T2 index 1

  Table T2(adjacency)

  Match: index select {size 3 : ECMP_1,ECMP_2,ECMP_3} ->removing ECMP_3 will rehash all traffic

- HW 4

  Table T1 (router)

  match : 192.168.4.0/24 action : go to table T2 index 1

  Table T2(adjacency)

  Match: index select {1K , ECMP_1*341,ECMP_2*341,ECMP_3*342}

  Table T2(adjacency) ->after removing ECMP_3 from route

  Match: index select {1K , ECMP_1*341,ECMP_2*341,ECMP_1*171, ECMP_2*171}

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Problem Statement- LPM example

- Linux pipeline

  192.168.1.0/24 via 172.17.42.1

  192.167.0.0/16 via 172.17.42.1

  192.0.0.0/8 via 172.17.42.1

  0.0.0.0/0 via 172.17.42.1

- HW 1 – TCAM base router

  Table T1 (router)

  match : 192.168.1.0 mask 255.255.255.0 action : go to table T2 index 1

  match : 192.167.0.0 mask 255.255.0.0 action : go to table T2 index 1

  match : 192.0.0.0 mask 255.0.0.0 action : go to table T2 index 1

  match : 0.0.0.0 mask 0.0.0.0 action : go to table T2 index 1

  Adding a /24 route may create a TCAM rule movement , may affect insertion rate

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Problem Statement- LPM example cont

- HW 2 – Hash base router

Table T1 (router /24)

match : 192.168.1.0 action : go to table T20 index 1

Default : go to T2

Table T2 (router /16)

match : 192.167.0.0 action : go to table T20 index 1

Default : go to T3

Table T3 (router /8)

match : 192.0.0.0 action : go to table T20 index 1

Default : go to T4

Table T4 (router /0)

Default : go to table T20 index 1

The structure of lookup tree and number prefixes of may affect PPS

.

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Proposed Solution

- HW pipe line visibility API (inspired by john fastabend flow APi)
- New devlink in order to :
  - Get device tables
  - Get device tables graph
  - Get all flows on a given table (issue #2)
  - Set table debug mode(add counters for table entry) ?
  - Get notification on flow add/remove (issue #1)

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Proposed Solution – example

sw:~$ sudo dl dev show tables

```
table  router
    match:
        metadata: VRF(ternary)
        fields:dst_ip  (ternary)
    action:
        set_forwording_action, set_metadata.ecmp_group
table ecmp_select
    match:
        metadata: ecmp_group(exact)
    action:
        ,set_metadata.adjacency(size,meta.ecmp_hash)
table  adjacency
    match:
        metadata: adjacency
    action:
        Set_feild.valn,set_feilds.dst_mac  ..
```

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Proposed Solution – example

sw:~$ sudo dl dev show table router

```
1  key {1(255),192.168.1.0(255.255.255.0) action {meta.ecmp_group=1} count 1000
1000  key {1(255),0.0.0.0(0.0.0.0) action {meta.ecmp_group=2} count 345
```

sw:~$ sudo dl dev show table ecmp_select

```
key 1 action {set_metadata.adjacency=0x10 +meta.ecmp_hash%4} count 1000
key 2 action {set_metadata.adjacency=0x20 +meta.ecmp_hash%2} count 345
```

sw:~$ sudo dl dev show table dadjacency

```
key 0x10 action {1,00:11:22:33:44:11} count 249
key 0x11 action {2,00:11:22:33:44:22} count 240
key 0x12 action {3,00:11:22:33:44:33} count 260
key 0x13 action {1,00:11:22:33:44:11} count 251
key 0x20 action {4,00:11:22:33:44:44} count 171
key 0x21 action {5,00:11:22:33:44:55} count 174
```

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Ordering Problems in Hardware Offloading

netdev 1.2

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Problem Statement

- Unlike in software forwarding, different layers need to be made aware of each other
- Problem is not specific to switchdev
  - Present in VLAN / MAC filtering
  - Solved by caching info on master device
  - Propagated to netdevs upon enslavement
- Example:
  - `$ ip link set dev team0 master br0`
  - `$ ip link set dev sw1p1 master team0`
  - Result: ASIC is not aware of the bridge device
  - “Correct” order is bottom-up

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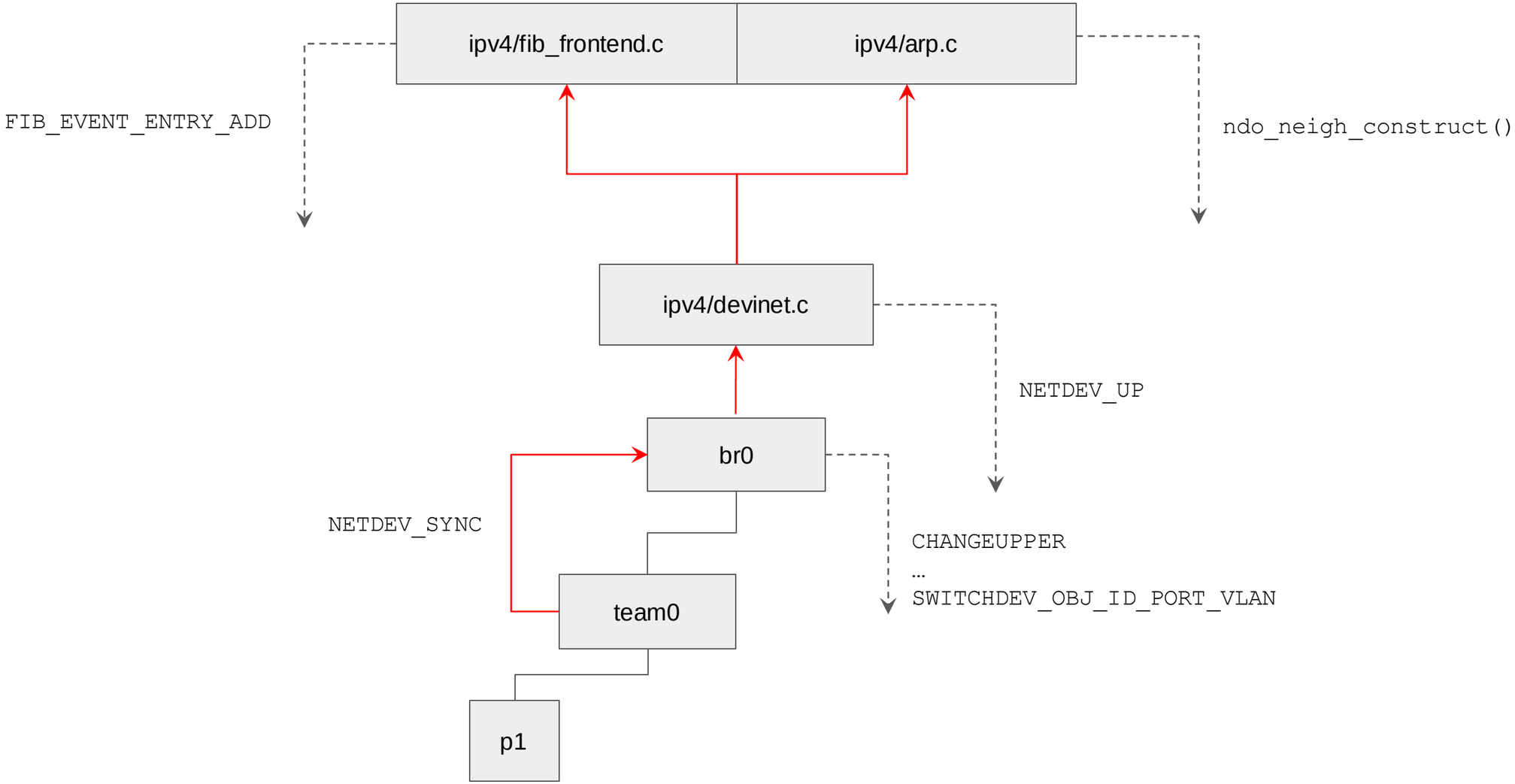
Proposed Solution

- Send an event when netdevs are added / removed as slaves of a master device (e.g., bond, team, bridge)
- Notification blocks can react and in turn relay the event up the stack
- Concept is already employed by bond and team to generate gratuitous ARP when slave configuration changes (NETDEV_NOTIFY_PEERS)
- Question: Should this be part of the netdev notification chain or switchdev's?

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Proposed Solution - Example

- Event chain upon enslavement of p1 to team0:



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Thank You

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